EE 354/CE 361/MATH 310 -

Introduction to Probability and Statistics

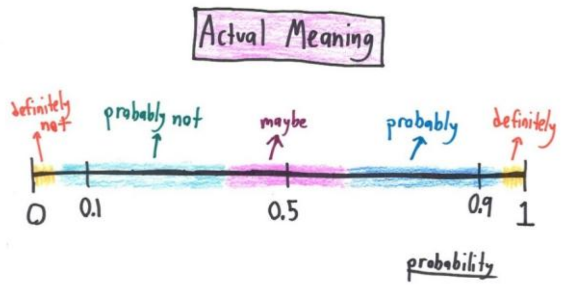
Spring 2026

EE 354/CE 361/MATH 310 L4 section (Spring 2026)

Introduction to Probability and Statistics -

Aamir Hasan

Week 4 - Lecture No 07 – February 02, 2026



Announcements!

1. Office hours – Tuesday & Thursday 12:00 to 1:00 PM @ Room C-211 (or through meeting request)
2. Quiz No 02 (Unit 2)– February 04, 2026
3. Quiz No 03 (Unit 3) – February 18, 2026
4. Assignment No 01 posted – due date February 10, 2026 (11:59 PM)
5. 03 TAs included in MATH 310 team (see CANVAS course site for details)

Agenda for today

1. Unit 2 - Conditional Probability and Bayes' Rule (to be completed)
2. Unit 3 – Counting (start)

TAs -

EE 354/CE 361/MATH 310 - Introduction to Probability and Statistics, Spring 2026

Meet Your TAs
Zain Naqi

About Me
I am a Computer Science Junior. My career interests include machine learning, deep learning, and reinforcement learning. Apart from academics, I do competitive programming and run a small e-commerce business. Mostly, you will find me in the library discussion area.

Ehsaas Hours
- Tue 9:30 - 11:30
- Thu 9:30 - 11:30

Contact Me
zno9224@st.habib.edu.pk or “Zain Naqi” on Teams

Meet Your TAs
Rayyan Shah

About Me
I am a Computer Science Junior. I am interested in bioinformatics, AI, and game development. Outside of academics, I also enjoy cooking and baking, and I always have some new recipe to recommend. You can usually find me in the library or tapal.

Ehsaas Hours
- Mon 4:00-5:00
- Wed 4:00-5:00
- Fri 3:30-4:30

Contact Me
rs08770@st.habib.edu.pk
Or “Rayyan Ali Shah” on Teams

Meet Your TAs
Hiba Aiman Ali

About Me
I am a Computer Engineering Junior with a minor in South Asian Music. My interests include bioinformatics, IoT, music and sustainability. I also sing as a part of the Habib University Orchestra. When I’m not in class, I’m most likely in the music room, singing or *trying to* learn an instrument.

Ehsaas Hours
- Mon 2:30 - 4:00
- Wed 2:30 - 4:00
- Fri 12:00-1:00

Contact Me
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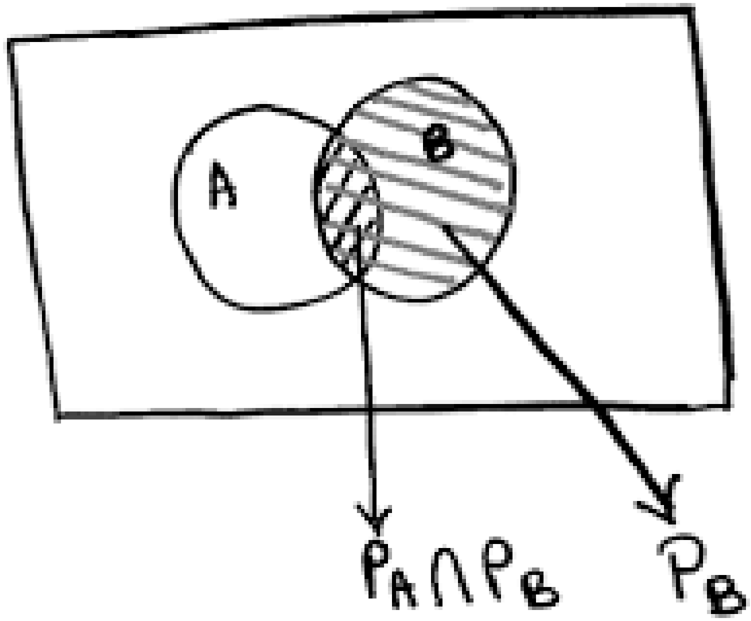
Unit 2: Conditional Probability and Bayes' Rule

- Conditional Probability
- Three important Tools
  - Multiplication rule
  - Total probability theorem
  - Bayes' rule (→ Inference)



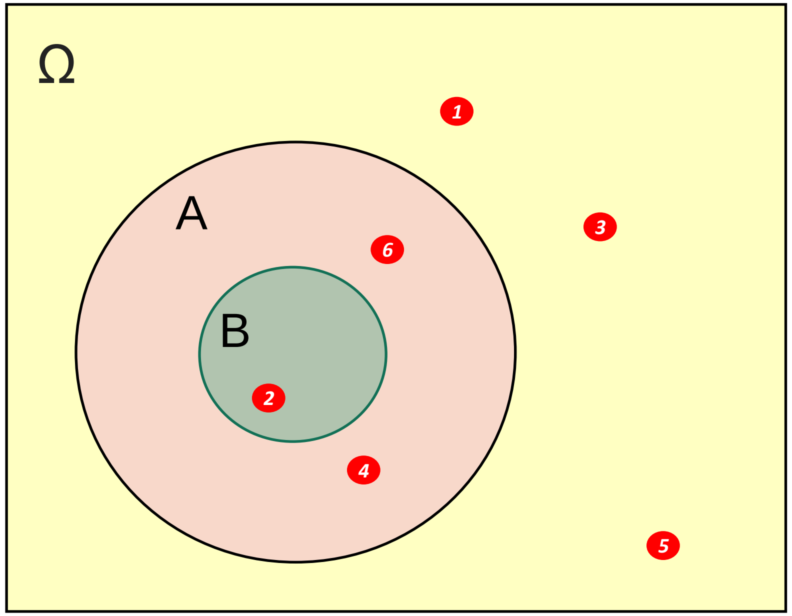
Consider a single roll of a fair dice (6 outcomes)

A = even outcome

B = even outcome < 4

C = odd outcome

$B \subseteq A$ Are A & B Independent?



$P(A) = \frac{3}{6}$

$P(B) = \frac{1}{6}$

$P(A|B) = ?$

$P(B|A) = ?$

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

defined only when $P(B) > 0$

Practice Problem 1 –

In a specified 6-AM-to-6-AM (d+1) 24-hour period, a student wakes up at time t_1 and goes to sleep at some later time t_2. Assume that 6AM at the start of cycle is 0, 7AM is 1 and so on, so that next day 6AM is time t = 24. Therefore, the sample space is given by: –

$$\Omega = \{(t_1, t_2) | t_1 \in [0, 24),\ t_2 \in (0, 24], \text{ and } t_2 > t_1\}$$

Note – (assume equally likely probability for wake up and sleep time)

Calculate the probability by specifying the **set A** and sketch the region on the plane corresponding to the event that the "*student is asleep at noon*".

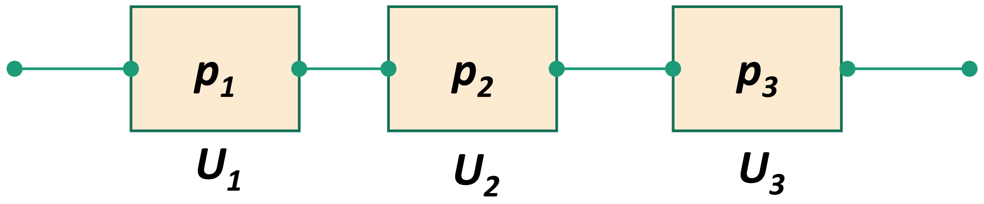
Application of Independence: Reliability (systems in Series)

- p_j: probability that unit j is "up"

Unit i is serviceable w.p p_i & denoted by U_i

Unit i is unserviceable w.p $(1 - p_i)$ & denoted by F_i

➢ $U_{i's}$ are independent, & therefore $F_{i's}$ are also independent



System is up if U_1, U_2, U_3 happens

➡ $$\mathrm{P}(\text{System is up}) = \mathrm{P}(U_1 \cap U_2 \cap U_3) = \mathrm{P}(U_1) \cdot \mathrm{P}(U_2) \cdot \mathrm{P}(U_3)$$

$$= p_1 \cdot p_2 \cdot p_3$$

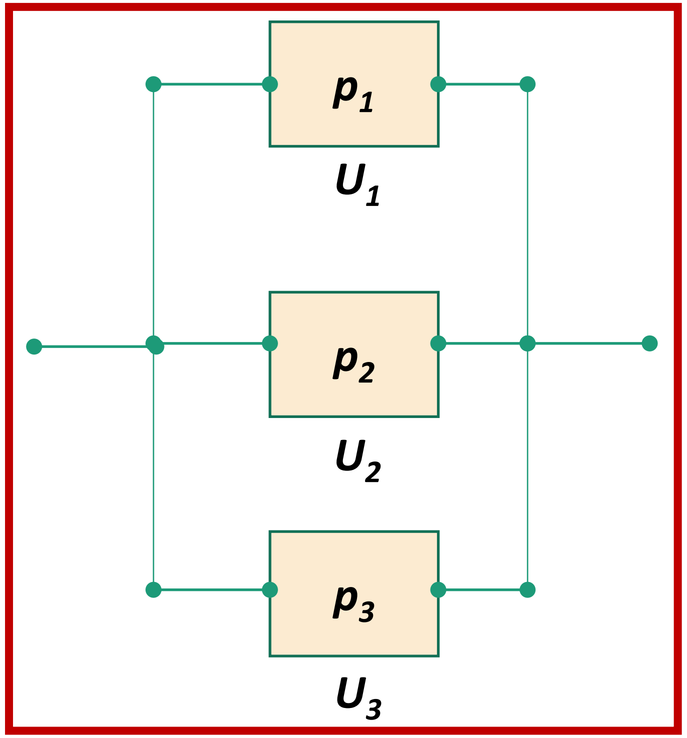
Application of Independence: Reliability (systems in Parallel)

- p_j: probability that unit j is "up"

Unit i is serviceable w.p p_i & denoted by U_i

Unit i is unserviceable w.p $(1 - p_i)$ & denoted by F_i

➢ *$U_{i's}$ are independent, & therefore $F_{i's}$ are also independent*



System is up if Path exists between A & B

$$\mathrm{P}(\text{System is up}) = \mathrm{P}(U_1 \cup U_2 \cup U_3)$$

$$= 1 - \mathrm{P}(U_1 \cup U_2 \cup U_3)^c$$

$$= 1 - \mathrm{P}(U_1^c \cap U_2^c \cap U_3^c)$$

$$= 1 - \mathrm{P}(F_1 \cap F_2 \cap F_3)$$

$$= 1 - \mathrm{P}(F_1) \,.\, \mathrm{P}(F_2) \,.\, \mathrm{P}(F_3)$$

$$= 1 - (1 - p_1) \,.\, (1 - p_2) \,.\, (1 - p_3)$$

Unit 2: Conditional Probability and Bayes' Rule

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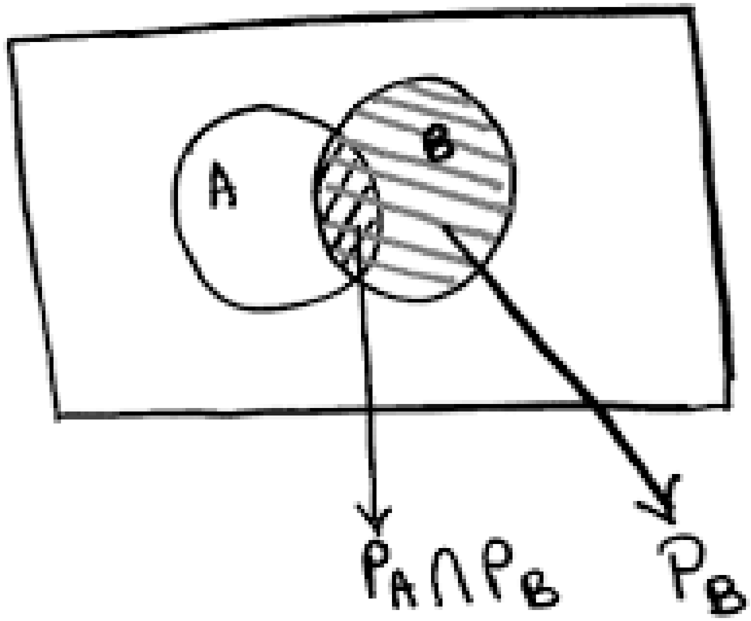
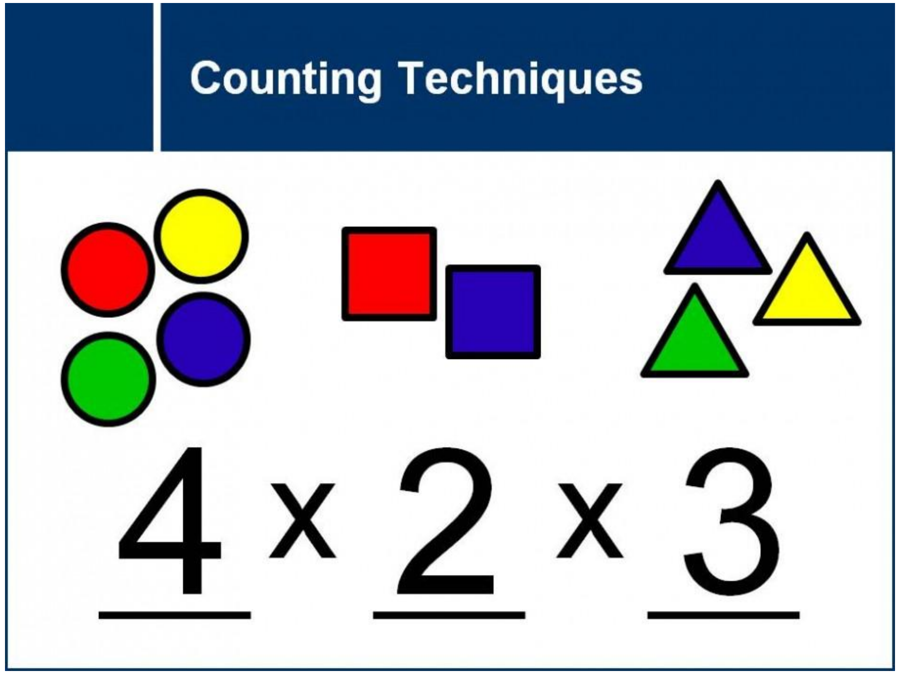


Unit 3: Counting

- Permutations
- Combinations

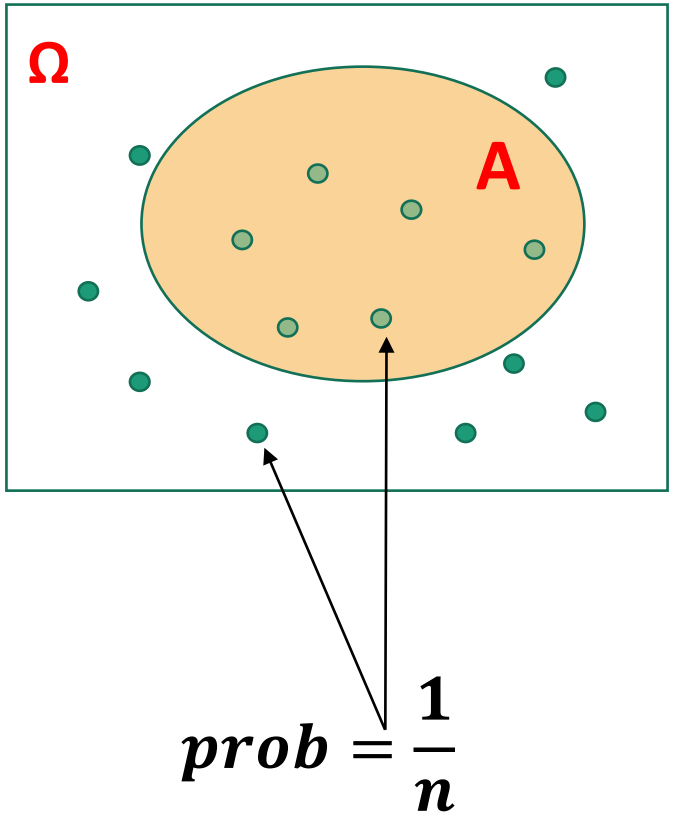


Counting – Why?

Discrete Uniform Law:

- Assume Ω consists of n equally likely elements
- Assume A consists of k elements

Then: $\mathbf{P}(A) = \dfrac{\textit{number of elements of } A}{\textit{number of elements of } \Omega} = \dfrac{k}{n}$



Practice Problem 1

"A certain class of 20 HU students are asked to undertake a class project comprising of two students. A geek HU programmer decides to write a Python program that lists all possible groups (N_1 = Number of possible groups) that can possibly be formed. She takes her program to a new level and based on some crude AI also predicts the group experience U_1 – based on student profiles and maps this on a scale of 3 (1-crappy, 2-ok, 3-excellent)"

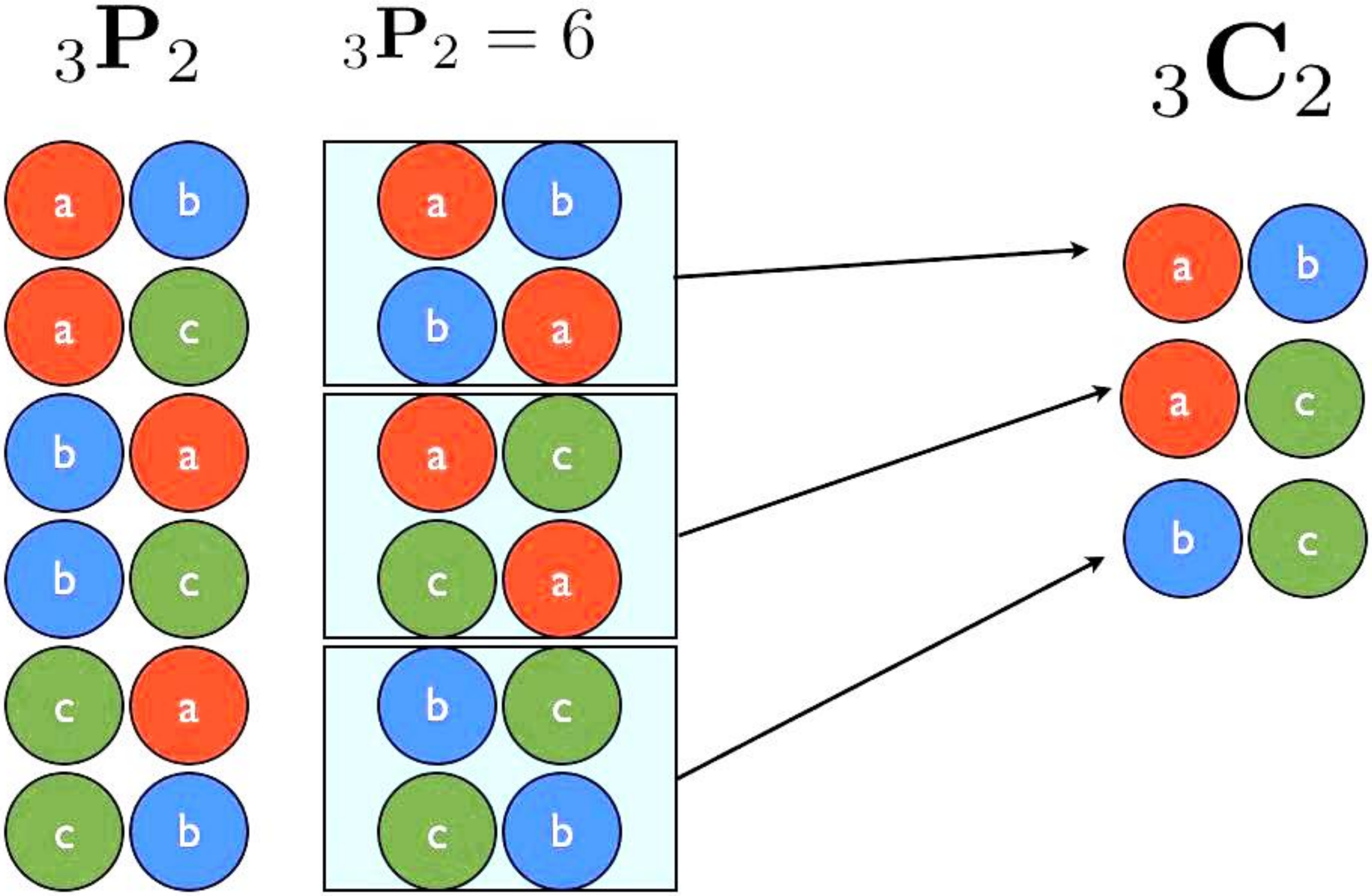
What is N_1?

"Based on a very poor feedback with regards to the class project exercise, in the next offering the instructor now asks each group to select a group lead & a member within each group. Assuming that this class also has 20 HU students, the number of possible group curations, N_2, are now mapped (using the same crappy AI) to predict the user experience U_2"

What is N_2?

After group activity – Discuss U_1 versus U_2

Counting



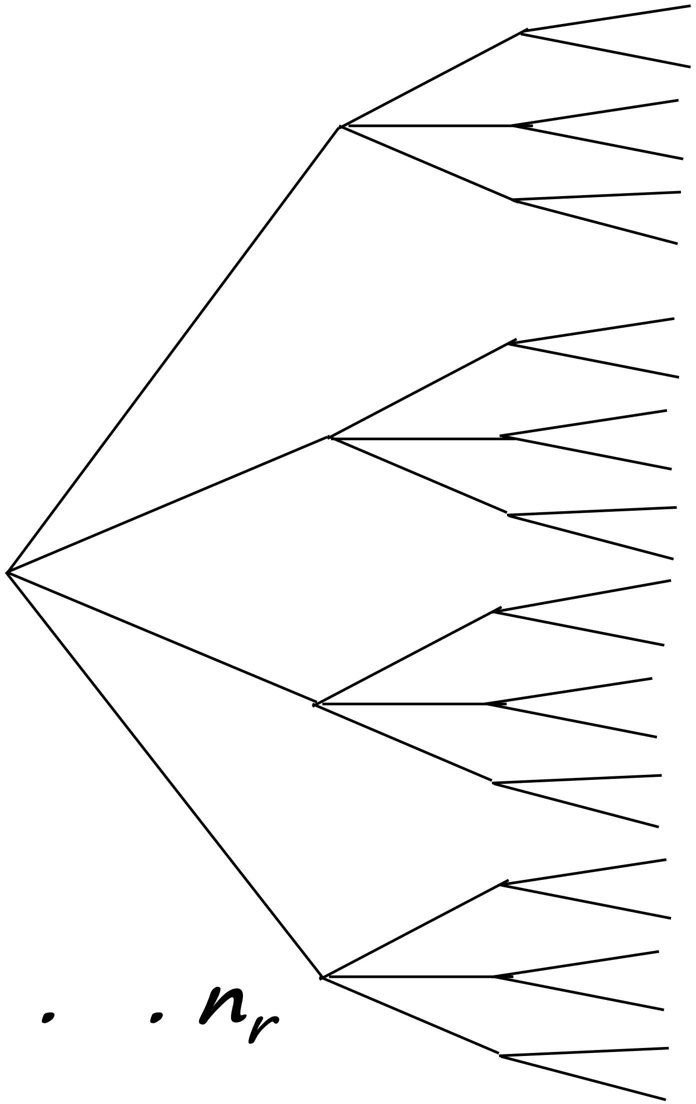
Basic Counting Principle

Example:

- 4 shirts.
- 3 ties.
- 2 jackets.

Number of possible attires?

4   12   24 $= 4 \times 3 \times 2$



$r = 3$

$n_1 = 4$

$n_2 = 3$

$n_3 = 2$

- r stages
- n_i choices at stage i

➡ *Number of choices is* $n_1 . n_2 . n_3 . \;.\;.\;. n_r$

Basic Counting Principle (ctd)

Example:

- **Number of license plates with 2 letters followed by 3 digits:**

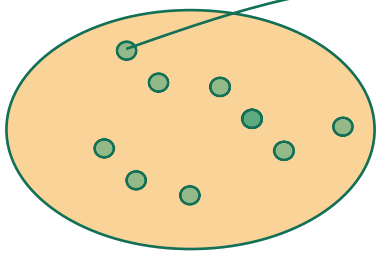
  ➡ $26 \cdot 26 \cdot 10 \cdot 10 \cdot 10$

  - What if repetition is prohibited?

    ➡ $26 \cdot 25 \cdot 10 \cdot 9 \cdot 8$

- **Permutations**: *Number of ways of ordering n elements*

  ➡ $= n \cdot (n-1) \cdot \cdot \cdot \cdot 1 = n!$

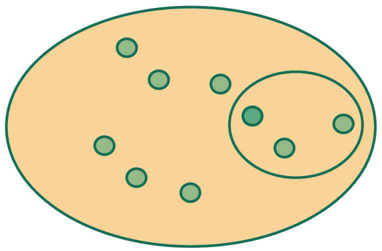


n	$n-1$	$n-2$	.	.		1

- **Number of distinct subsets of {1, … , n}:** $= 2^n$

Remember this!

?



2	2	2	.	.		2

Example

- Find the probability that:
  six rolls of a (six sided) die all give different numbers.
  (Assume all outcomes equally likely.)

A

➡ Typical outcome: $P(2, 3, 4, 5, 6, 2) = \frac{1}{6^6}$

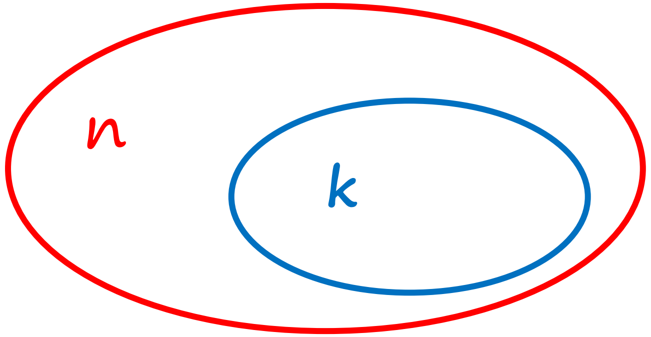
➡ Typical element of A: (2, 3, 4, 1, 6, 5) = 6!

$$P(A) = \frac{\#\, in\, A}{\#\, possible\ outcomes} = \frac{6!}{6^6}$$

Combinations

Definition: $\binom{n}{k}$ Number of k-element subsets of a given n-element set

$= \binom{n}{k} = \frac{n!}{k!(n-k)!}$



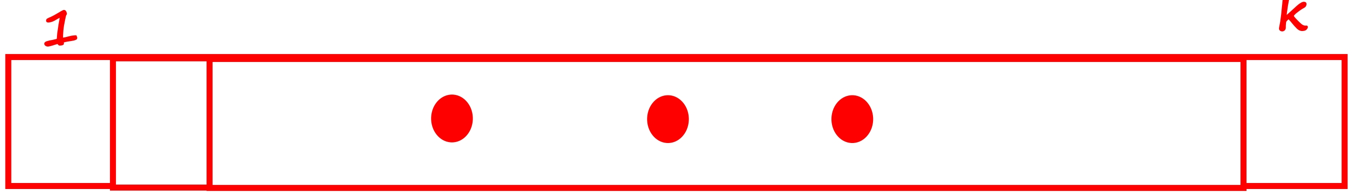
➡ $n = 0, 1, 2, \ldots$

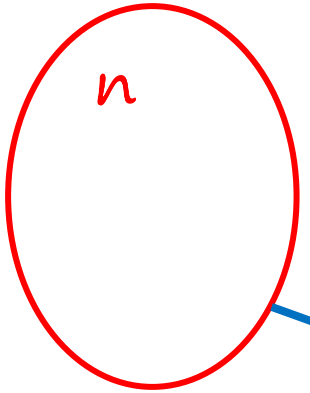
➡ $k = 0, 1, 2, \ldots, m$

- Two ways of constructing an **ordered** sequence of *k* **distinct** items:

  - Option 1 - Choose the ***k*** items one at a time.

  - Option 2 - First Choose ***k*** items, then order the ***k*** them.

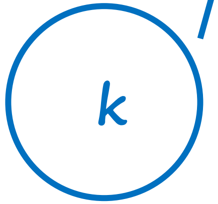




➡ $n(n-1)(n-2)\ldots(n-k+1) = \frac{\mathbf{n}!}{(\mathbf{n}-\mathbf{k})!}$

$k!$

$\binom{n}{k}$



$\binom{n}{k}k!$

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}$$

$\binom{n}{n} = 1 \qquad \frac{n!}{n!\,0!} \qquad 0! = 1$ convention

$\binom{n}{0} = \frac{n!}{0!\,n!} = 1 \qquad \emptyset$

$\sum_{k=0}^{n} \binom{n}{k} = \binom{n}{0} + \binom{n}{1} + \ldots \binom{n}{n} = \#\ all\ subsets = 2^n$

Binomial coefficient $\binom{n}{k}$ → Binomial probabilities

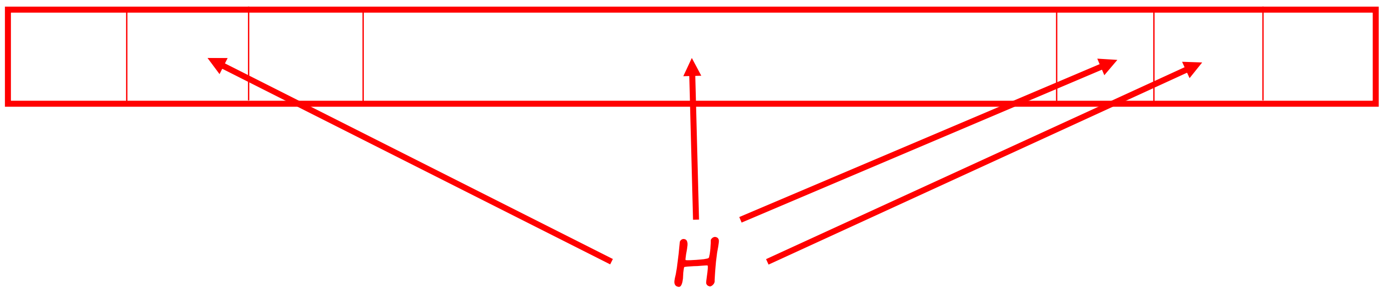
$$\mathbf{P}\,(k \text{ heads}) = \binom{n}{k} p^k (1-p)^{n-k}$$

- n >= 1 independent coin tosses; $\mathbf{P}\,(H) = p$

n=6

- $\mathbf{P}\,(H\,T\,T\,H\,H\,H) = P\,(1-p)\,(1-p)\,p\;p\;p = p^4\,(1-p)^2$
- $\mathbf{P}\,(\textit{particular sequence}) = p^{\#heads}\,(1-p)^{\#tails}$
- $\mathbf{P}\,(\textit{particular k-head sequence}) = p^k\,(1-p)^{n-k}$

➡ $\mathbf{P}\,(k \text{ heads}) = p^k\,(1-p)^{n-k}$ * (# of k-head sequence)



$$\binom{n}{k}$$

A coin tossing problem

Assumptions:

- **independence**
- $\mathbf{P}(H) = p$

$$\mathbf{P}\,(k \text{ heads}) = \binom{n}{k} p^k (1-p)^{n-k}$$

- Given that there were 3 heads in 10 tosses, what is the probability that the first two tosses were heads?
  - event *A* : the first 2 tosses were heads.
  - event *B* : 3 out of 10 tosses were heads.